NetSage AI – Individual Project Summary

Cisco-AICTE VIP 2026 | AI Track

1. Project Overview

NetSage AI is an AI-assisted network troubleshooting project designed to diagnose common network configuration problems using topology details, symptoms, and command-line evidence. The workflow combines AI diagnosis with human review and independent rule-based verification.

2. Objectives

• Identify the root cause of network connectivity and configuration issues.
• Use network evidence for structured diagnosis.
• Generate an AI diagnosis with the affected concept and suggested fix.
• Compare AI results with human review as Accepted, Edited, or Rejected.
• Validate selected faults using Cisco Packet Tracer and Python rules.
• Present project results through an Excel dashboard.

3. Dataset and Issue Coverage

The project contains 30 network troubleshooting cases covering VLAN (4), Routing (8), DHCP (4), DNS (2), ACL (5), NAT (3), Wireless (3), and Gateway (1).

4. AI Diagnosis and Human Review

Each case was reviewed by comparing the AI output with the available network evidence. The final tracker records 16 Accepted, 8 Edited, and 6 Rejected outcomes.

5. Severity Analysis

The final dashboard contains 7 Critical, 14 High, and 9 Medium cases.

6. Cisco Packet Tracer Demonstration

The demonstration uses a VLAN mismatch scenario. PC0 (192.168.10.10/24) and PC1 (192.168.10.11/24) initially could not communicate. Switch verification showed Fa0/2 in VLAN 10 and Fa0/3 in VLAN 20. Fa0/3 was assigned to VLAN 10. After the correction, the ping succeeded with 0% packet loss.

7. Deterministic Rule-Based Verification

A Python rule checker independently validates network evidence using predefined rules. This provides an additional verification layer instead of depending only on the AI diagnosis.

8. Dashboard and Analysis

The final Excel dashboard presents Issue Type Distribution, Severity Distribution, and AI vs Human Review outcomes.

9. My Project Contribution

My contribution focused on Packet Tracer implementation, VLAN troubleshooting demonstration, dashboard preparation, and demonstration video recording.

10. Conclusion

NetSage AI demonstrates a practical workflow for AI-assisted network troubleshooting. Human review, Packet Tracer validation, and deterministic rule-based checking provide verification beyond the AI output. The overall workflow is: Diagnose → Review → Fix → Verify.